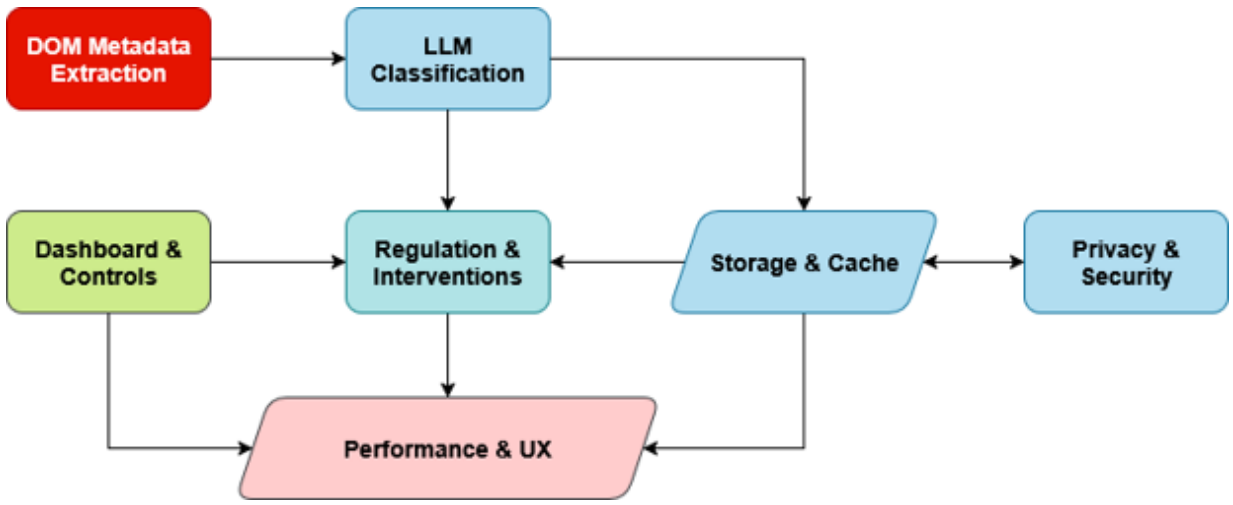
Project Acceptance Tests

Description

This document defines various acceptance tests covering the end-to-end functionality required by the **Scope of Work** and **User Stories**. The tests will validate: accurate LLM-based video classification; reliable metadata extraction; configurable regulation of content (including allostatic function); dashboard visualisation; interventions (blur/hide/block); performance; privacy and local-first storage; testability/maintainability via unit-testing.

Testing Strategy

- **Subsystems:** (1) DOM Metadata Extraction, (2) LLM classification, (3) regulation & interventions, (4) dashboard & controls, (5) storage & cache, (6) performance & UX, (7) privacy & security.
- **Integration:** Bottom-up unit testing of individual subsystems → integration testing on Youtube pages → automated testing using Selenium/Playwright → client acceptance walkthrough.



Tests

Test A - Video Classification Accuracy (LLM)

Validates that the LLM classifies videos into agreed categories with high accuracy, enabling user control and regulation.

Test Specification

- **Requirements covered**: LLM classifier; accuracy target; client satisfaction.
- **Success criteria:** >95% on 100-1000 sample set
- **Method:** one/few shot prompt; cached classification call.

Test Description

- **Means of Control:** Automated testing; no retries → record mis-classifications.
- **Data:**
    - Input Data → testing video metadata set (title, description, tags, label) → scraped from YouTube API
    - Output Data → csv of video title predicitons, test metric report
    - System Messages → summary log & table of misclassification table
- **Procedures:** run test script → verify results → archive test report

Test Analysis Report

- **Function:** correct category assignement
- **Performance:** average response <1000ms
- **Outcome**: pass/fail → if fail, revision of prompt and/or model selection.

Test B - Metadata Extraction

Ensures title/description/tags/video ID are reliably captured for each visible video card.

Test Specification

- **Requirements covered**: Ensure a metadata is retrieved for each video so content can be accurately categorised.
- **Success criteria**: >99% of visible cards have non-empty title and ID; description/tags captured when present.

Test Description

- **Means of Control**: Playwright/selenium loads live YouTube page.
- **Data:**
    - **Input Data**: live youtube pages/curated youtube playlists.
    - **Output Data**: JSON of extracted information relating to each video
    - **System Messages**: alerts of missing fields that are expected
- **Procedures:** Scroll to load ~200 cards; assert metadata scrape; export scrape summary.

Test Analysis Report

- **Function:** DOM scrape correctness
- **Performance:** extraction time per batch of videos
- **Data measures:** complete fields %, error counts.
- **Outcome**: Pass if test thresholds met; else fix selectors and retry.

Test C - Regulation Controls (for each category & level)

Validates sliders/dials per category and the resulting intervention.

Test Specification

- **Requirements covered**: User can set regulation levels; gentle interventions; not breaking layouts.
- **Success criteria**: Changing a slider updates behaviour on without refresh (allows for live assessment).

Test Description

- **Means of Control**: Manual testing, visually assessing page changes
- **Data**
    - **Input Data**: Preset categories; test tiles tagged by genre.
    - **Input Commands**: Adjust sliders; toggle blur/hide mode.
    - **Output Data**: DOM state per tile (visible/blurred/removed).
    - **System Messages**: message for mode changes.
- **Procedures:** For each category, set level High, Medium and Low → assess changes to DOM

Test Analysis Report

- **Function**: Correct mapping of level → intervention.
- **Performance**: DOM updates immediately
- **Data measures**: % tiles matching expected state.

Test D - Allostatic Regulation & Time-Series Graph

Test Specification

- **Requirements covered**: Allostatic mode; time-series graph per category; user comprehension.
- **Success criteria**: Exposure up → regulation level increases; graph reflects values.

Test Description

- **Means of Control**: Simulated exposure stream; dashboard open.
- **Data**
    - **Input Data**: numerous events in one category → assess allosatic function and regulation behaviour if threshold is passed.
    - **Output Data**: Regulation values; chart points.
    - **System Messages**: None beyond debug logs.
- **Procedures**: Feed events; assert penalty rises; assert decay after cooldown following allostatic function.

Test Analysis Report

- **Function**: decay; chart accuracy.
- **Performance**: Very low update latency per event.
- **Data measures**: penalty vs event count; chart–data parity test.
- **Outcome**: Pass if decay rules and chart parity hold.

Test E - Dashboard & Control Panel UX

Validates that users can understand and operate the interface without maths knowledge.

Test Specification

- **Requirements covered**: Easy-to-understand interface + analytics; toolbar active icon; extension toolbar popup;
- **Success criteria**: SUS (System Usability Score) ≥ 80 in quick usability survey; all controls discoverable.

Test Description

- **Means of Control**: Moderated usability session (5 or so participants.

- **Data**
    - **Input Data**: Tasks list (change a slider, add category, find graph, toggle dark mode).
    - **Input Commands**: Click through UI; complete survey.
    - **Output Data**: SUS scores; event logs.
    - **System Messages**: Onboarding tooltips appear contextually.
- **Procedures**: Run script; collect SUS; verify automated visibility checks.

Test Analysis Report

- **Function**: Feature discoverability & comprehension.
- **Data measures**: SUS score; task completion rate/time.
- **Outcome**: Pass if SUS ≥ 80 and 100% task completion.

Test F - Performance & Page Load Budget (optional optimisation if time allows).

Ensures the extension does not degrade YouTube experience.

Test Specification

- **Requirements covered**: YouTube page loads; caching; responsiveness.
- **Success criteria**: DOM content loads; minimal CPU overhead.
- **Means of Control**: Playwright/selenium; cold and warm cache runs.
- **Data**
    - **Input Data**: Pre-seeded cache for warm runs.
    - **Output Data**: Trace JSON; summary CSV.
    - **System Messages**: Perf budget results.
- **Procedures**: Run cold; run warm; compare metrics; attach results.

Test Analysis Report

- **Function**: Acceptable load and scroll performance.
- **Performance**: As above.
- **Data measures**: long tasks, CPU time slice.
- **Outcome**: Pass if all budgets met.

Test G - Storage & Cache

Validates that, after a video is classified, the result is persisted to the cache and is served on subsequent encounters (i.e., hit on revisit/scroll), including versioning and expiry rules.

Test Specification

- **Requirements covered**: Uses local cache to improve user experience + prevent storage issues.
- **Success criteria**:
    1. On first classify (miss), entry is **written** to cache
    2. On revisit (hit), classification is **served from cache** without calling the LLM.
    3. Cached result must match last successful classification.
    4. Changing **model version** or **prompt version** results in a miss + refresh write.
    5. Entries **expire** after time or storage limit reached.
    6. Under concurrent requests for the same video ID, only one entry

Test Description

- **Means of Control**: Network inspector to assert zero LLM calls on cache hits.
- **Input Data**: 30 known video IDs, fixed model version, TTL (time to live) of 30 days.
- **Procedure**:
    - Cold run (no cached classification): expect LLM call and cache write.
    - Warm run (videos already cached): expect cache hit and no LLM call.
    - Version change: expect LLM call and cache update.
    - Data expiry: expect refresh from LLM.
- **Expected Output**: Cache entries created, hits on revisit, refresh on version/expiry.

Test Analysis Report

- **Function**: Cache stores classification, retrieves correctly, refreshes when needed.
- **Performance**: Immediate warm load classification retrieval.
- **Pass Criteria**: 100% hits for unchanged videos; refreshes correctly on version/expiry.

Test Materials

- **Datasets**: 100-1000 item labelled set.
- **Tools**: Playwright, Selenium.
- **Environments**: Chrome on Windows/macOS; extension dev build (MV3).
- **Documentation**: Onboarding script for client UAT; UX survey (SUS) form; this acceptance test plan.